

2. Bayesian Inference

- 1 Bayesian Models
- 2 MLE
- 3 MAP
- 4 Some Practice Problems

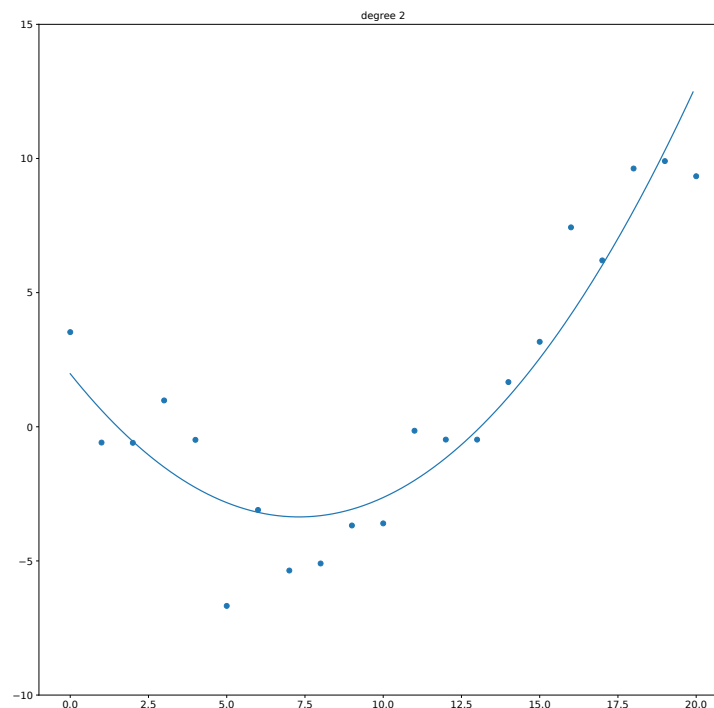
Outline

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Motivation

- Recall:

- ▶ Classification, regression, prediction, etc.: $\mathcal{D} = \{(\mathbf{x}_i, y_i) : i = 1, \dots, n\}$
- ▶ Fitting a quadratic function $y = a_0 + a_1x + a_2x^2$ to training samples (x_i, y_i) .
“Optimal” values of a_0, a_1, a_2 ?



- Typical goal: learn a probability distribution that generates $\mathcal{D}$.

Examples of Likelihood models

Training data set $\mathcal{D} = \{x_1, \dots, x_n\}$:

- $x_i \sim \text{Bern}(\cdot | \theta)$ i.i.d., where $\theta \in [0, 1]$, then

$$p(\mathcal{D} | \theta) = \theta^{N_1} (1 - \theta)^{N_0}, \quad N_k = \sum_{i=1}^n \mathbf{1}\{x_i = k\}.$$

- The probability of the data measures how well the model fits the data.
- This is called the **likelihood** function (of θ with $\mathcal{D}$ fixed).
- E.g.: $\mathcal{D} = \{0, 1, 1, 0, 1, 1, 1, 0, 1, 1\}$, $N_1 = 7$, $N_0 = 3$.

$$p(\mathcal{D} | \theta) = \theta^7 (1 - \theta)^3.$$

- ▶ If $\theta = 0.6$, then $p(\mathcal{D} | \theta) = 0.6^7 (1 - 0.6)^3 = 0.6^7 \cdot 0.4^3$.
- ▶ This tells us how “likely” $\theta = 0.6$ is to have observed $\mathcal{D}$.
- ▶ Varying levels of likelihood for different θ .
- ▶ θ is unknown *a priori* and we want to infer it from the training data $\mathcal{D}$.

Examples of Likelihood models

Training data set $\mathcal{D} = \{x_1, \dots, x_n\}$:

- $x_i \sim \text{Exp}(\cdot | \lambda)$ i.i.d., where $\lambda > 0$, then

$$p(\mathcal{D} | \lambda) = \lambda^n \exp\left(-\lambda \sum_{i=1}^n x_i\right)$$

- E.g.: $\mathcal{D} = \{0.15, 0.28, 0.23, 0.14, 0.35\}$, $n = 5$.

- ▶ The likelihood of λ is

$$p(\mathcal{D} | \lambda) = \lambda^5 \exp(-1.15\lambda).$$

- ▶ Different values of λ give different likelihood values.
- ▶ Which value of λ is most likely to have generated the data?

Examples of Likelihood models

Training data set $\mathcal{D} = \{x_1, \dots, x_n\}$:

- $x_i \sim \mathcal{N}(\cdot | \mu, \sigma^2)$ i.i.d., then $\boldsymbol{\theta} = (\mu, \sigma^2)$ and

$$p(\mathcal{D} | \boldsymbol{\theta}) = \frac{1}{(2\pi)^{n/2} \sigma^n} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

- E.g.: $\mathcal{D} = \{0.24, -0.35, -0.16, 0.12\}$, $n = 4$.

► The likelihood of $\boldsymbol{\theta}$ is

$$p(\mathcal{D} | \boldsymbol{\theta}) = \frac{1}{(2\pi)^2 \sigma^4} \exp\left\{-\frac{1}{2\sigma^2} \left((0.24 - \mu)^2 + (0.35 + \mu)^2 + (0.16 + \mu)^2 + (0.12 - \mu)^2\right)\right\}.$$

$\boldsymbol{\theta} = (\mu, \sigma)$	$p(\mathcal{D} \boldsymbol{\theta})$
$(0, 1)$	0.0227
$(-1, 1)$	0.0036

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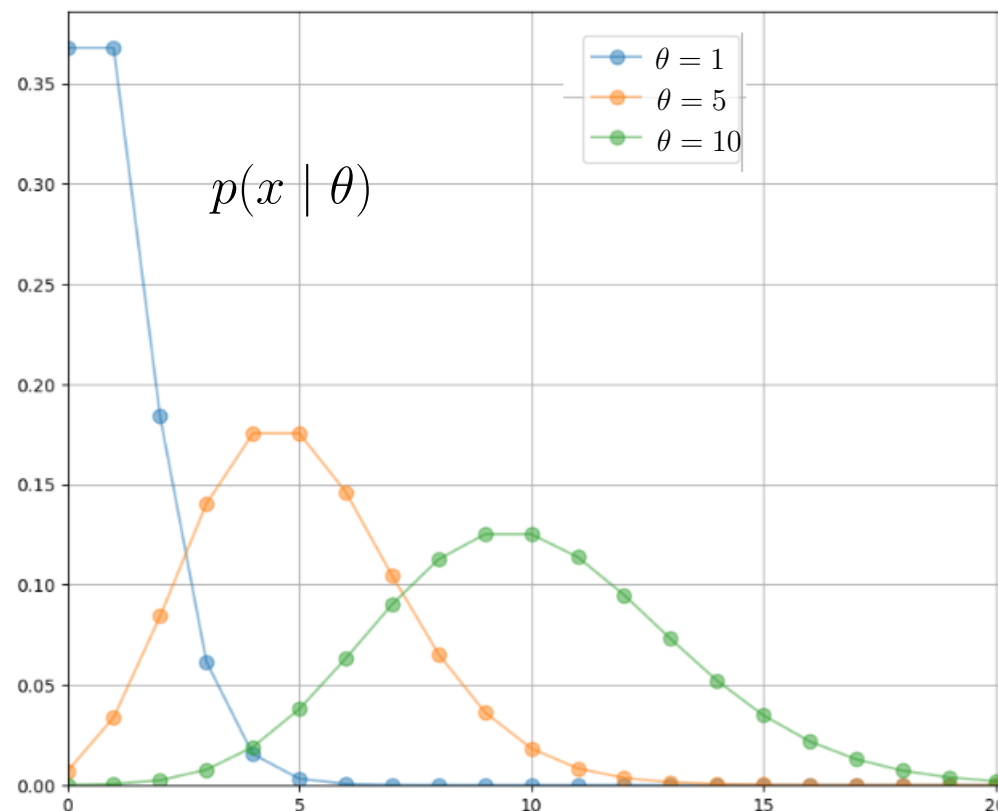
4 Some Practice Problems

Maximum Likelihood Estimate (MLE)

- MLE of a model's parameter θ :

$$\theta_{\text{ML}} = \arg \max_{\theta} \underbrace{p(\mathcal{D} \mid \theta)}_{\text{likelihood}}$$

- An observation or data $x \sim p(\cdot \mid \theta)$:



Maximum Likelihood Estimate (MLE)

- Given i.i.d. data $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, which $p(\cdot \mid \boldsymbol{\theta})$ is most “likely”?
- MLE: find $\boldsymbol{\theta}$ that maximizes the **log likelihood**:

$$\log p(\mathcal{D} \mid \boldsymbol{\theta}) = \log \prod_{i=1}^n p(\mathbf{x}_i \mid \boldsymbol{\theta}) = \sum_{i=1}^n \log p(\mathbf{x}_i \mid \boldsymbol{\theta}).$$

[Can you see why it is ok to use log? This is often done to simplify the expression.]

MLE Examples

- $p(\mathcal{D} \mid \theta) = \theta^{N_1} (1 - \theta)^{N_0}$, where $N_k = \sum_{i=1}^n \mathbf{1}\{x_i = k\}$.

$$\log p(\mathcal{D} \mid \theta) = N_1 \log \theta + N_0 \log(1 - \theta).$$

Taking derivatives w.r.t. θ and setting to zero:

$$\begin{aligned} \frac{N_1}{\theta_{\text{ML}}} &= \frac{N_0}{1 - \theta_{\text{ML}}} \\ \theta_{\text{ML}} &= \frac{N_1}{N_0 + N_1} = \frac{N_1}{n}. \end{aligned}$$

- E.g.: $\mathcal{D} = \{0, 1, 1, 0, 1, 1, 1, 0, 1, 1\}$, $N_1 = 7$, $N_0 = 3 \implies \theta_{\text{ML}} = 0.7$.

MLE Examples

- $x_i \sim \text{Exp}(\cdot | \lambda)$ i.i.d., where $\lambda > 0$.

$$\log p(\mathcal{D} | \theta) = n \log \lambda - \lambda \sum_{i=1}^n x_i.$$

Taking derivatives w.r.t. θ and setting to zero:

$$\frac{n}{\lambda_{\text{ML}}} = \sum_{i=1}^n x_i$$
$$\lambda_{\text{ML}} = \frac{n}{\sum_{i=1}^n x_i}.$$

- E.g.: $\mathcal{D} = \{0.15, 0.28, 0.23, 0.14, 0.35\}$, $n = 5$
 $\implies \lambda_{\text{ML}} = \frac{5}{0.15+0.28+0.23+0.14+0.35} = \frac{5}{1.15}.$

Kullback-Leibler Divergence (Supplementary)

- **Strong Law of Large Numbers (SLLN)**: Suppose $X_1, X_2, \dots$ are i.i.d. random variables. Then, with probability one,

$$\frac{1}{n} \sum_{i=1}^n X_i \rightarrow \mathbb{E}[X_1]$$

as $n \rightarrow \infty$.

- Suppose that θ_0 is the actual parameter value.
- When n is large, from the **SLLN**,

$$\frac{1}{n} \sum_{i=1}^n \log p(\mathbf{x}_i \mid \theta) \approx \mathbb{E}_{\theta_0}[\log p(\mathbf{x} \mid \theta)]$$

- Therefore, MLE is approximately **minimizing**

$$-\mathbb{E}_{\theta_0}[\log p(\mathbf{x} \mid \theta)] + \underbrace{\mathbb{E}_{\theta_0}[\log p(\mathbf{x} \mid \theta_0)]}_{\text{const.}} = \mathbb{E}_{\theta_0} \left[\log \frac{p(\mathbf{x} \mid \theta_0)}{p(\mathbf{x} \mid \theta)} \right] \triangleq D(p_{\theta_0} \parallel p_{\theta}).$$

- $D(p_{\theta_0} \parallel p_{\theta})$ is the **Kullback-Leibler divergence** between p_{θ_0} and p_{θ} . It measures how “different” these two distributions are.

Kullback-Leibler Divergence (Supplementary)

- Shannon (1948): $D(p \parallel q) \geq 0$ with equality iff $p(x) = q(x)$ with probability one under both p and q .

Proof.

Jensen's inequality: If f is convex, $\mathbb{E}|X| < \infty$, and $\mathbb{E}|f(X)| < \infty$, then $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$. Furthermore, if f is strictly convex, then equality holds iff X is a constant.

$$\begin{aligned} D(p \parallel q) &= \mathbb{E}_p \left[\log \frac{p(x)}{q(x)} \right] = -\mathbb{E}_p \left[\log \frac{q(x)}{p(x)} \right] \geq -\log \mathbb{E}_p \left[\frac{q(x)}{p(x)} \right] \\ &= -\log \int \frac{q(x)}{p(x)} p(x) \, dx = 0, \end{aligned}$$

where equality holds iff $\frac{q(x)}{p(x)} = \text{const}$, i.e., $p(x) = q(x)$ for all x with probability one. □

- MLE is approximately minimizing $D(p_{\theta_0} \parallel p_{\theta})$ when n is large.
- MLE tries to find the parameter that makes p_{θ} as “close” to p_{θ_0} as possible.

Example: Cross-entropy Loss

- Cross-entropy loss is widely used in classification problems.
- Suppose we have K classes and the true class label of a data point $\mathbf{x}$ is $y \in \{1, \dots, K\}$.
- To predict the class label, we can use a model $q(\cdot \mid \mathbf{x}, \boldsymbol{\theta})$ that outputs a probability distribution over the K classes.
- To find the best parameter $\boldsymbol{\theta}$, we can use MLE: maximize the log likelihood of the training data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$:

$$\begin{aligned}
 \sum_{i=1}^n \log p(\mathbf{x}_i, y_i \mid \boldsymbol{\theta}) &= \sum_{i=1}^n \log q(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}) + \sum_{i=1}^n \log p(\mathbf{x}_i \mid \boldsymbol{\theta}) \\
 &= \sum_{i=1}^n \sum_{k=1}^K \mathbf{1}\{y_i = k\} \log q(k \mid \mathbf{x}_i, \boldsymbol{\theta}) + \sum_{i=1}^n \log p(\mathbf{x}_i) \\
 &= -n \cdot \underbrace{\left(-\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \mathbf{1}\{y_i = k\} \log q(k \mid \mathbf{x}_i, \boldsymbol{\theta}) \right)}_{\text{cross-entropy loss}} + \text{const},
 \end{aligned}$$

i.e., we minimize the **cross-entropy loss**.

Example: Linear Regression

- y - response variable depends on input variables $\mathbf{x} = (x_1, x_2, \dots, x_D)$ in a linear model:

$$y = \mathbf{w}^\top \mathbf{x} + \epsilon,$$

where ϵ is Gaussian $\mathcal{N}(0, \sigma^2)$, then $p(y \mid \mathbf{x}, \mathbf{w}) = \mathcal{N}(y \mid \mathbf{w}^\top \mathbf{x}, \sigma^2)$.

- Suppose we have i.i.d. training data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$:

$$\begin{aligned} \text{log likelihood} \quad \log p(\mathbf{y} \mid \phi(\mathbf{x}), \mathbf{w}) &= \sum_{i=1}^n \log \mathcal{N}(y_i \mid \mathbf{w}^\top \mathbf{x}_i, \sigma^2) \\ &= -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + \text{const} \\ &= -\frac{1}{2\sigma^2} \|\Phi \mathbf{w} - \mathbf{y}\|^2 + \text{const} \end{aligned}$$

where $\Phi = [\mathbf{x}_1 \cdots \mathbf{x}_n]^\top$. MLE of $\mathbf{w}$:

$$\min_{\mathbf{w}} \|\Phi \mathbf{w} - \mathbf{y}\|^2$$

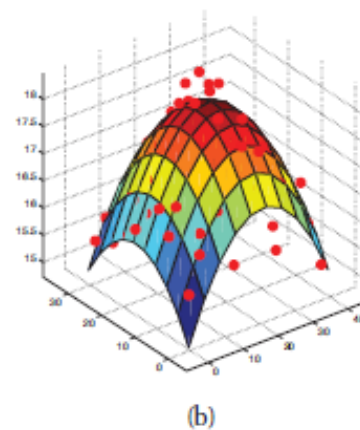
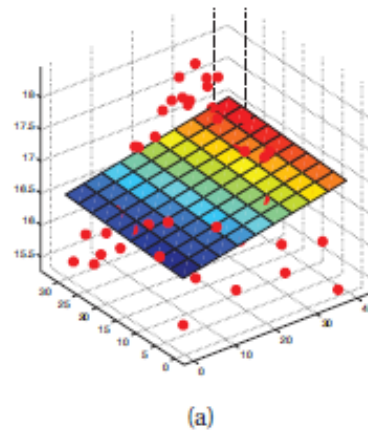
Example: Linear Regression

- Model non-linear relationships using **basis function** expansion. Choose basis functions $\varphi_i : \mathbb{R}^D \rightarrow \mathbb{R}$, $i = 1, \dots, M$.

$$\phi(\mathbf{x}) = \begin{bmatrix} \varphi_1(\mathbf{x}) \\ \varphi_2(\mathbf{x}) \\ \dots \\ \varphi_M(\mathbf{x}) \end{bmatrix}, \quad y = \mathbf{w}^\top \phi(\mathbf{x}) + \epsilon$$
$$= \sum_{i=1}^M w_i \varphi_i(\mathbf{x}) + \epsilon \sim \mathcal{N}(\cdot \mid \mathbf{w}^\top \phi(\mathbf{x}), \sigma^2)$$

Linear Basis Function Examples

- $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.
- $\phi(\mathbf{x}) = [1, x_1, x_2, x_1^2, x_2^2] \implies y = w_1 + w_2x_1 + w_3x_2 + w_4x_1^2 + w_5x_2^2 + \epsilon$.



Linear regression applied to 2d data. Vertical axis is temperature, horizontal axes are location within a room. Data was collected by some remote sensing motes at Intel's lab in Berkeley, CA (data courtesy of Romain Thibaux). (a) The fitted plane has the form $\hat{f}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2$. (b) Temperature data is fitted with a quadratic of the form $\hat{f}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2$.

MLE of the Weights

- Suppose we have i.i.d. training data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$:

log likelihood

$$\begin{aligned}
 \log p(\mathbf{y} \mid \phi(\mathbf{x}), \mathbf{w}) &= \sum_{i=1}^n \log \mathcal{N}(y_i \mid \mathbf{w}^\top \phi(\mathbf{x}_i), \sigma^2) \\
 &= -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{w}^\top \phi(\mathbf{x}_i))^2 + \text{const} \\
 &= -\frac{1}{2\sigma^2} \|\Phi \mathbf{w} - \mathbf{y}\|^2 + \text{const} \\
 &= -\frac{1}{2\sigma^2} (\mathbf{w}^\top \Phi^\top \Phi \mathbf{w} - 2\mathbf{y}^\top \Phi \mathbf{w}) + \text{const},
 \end{aligned}$$

where $\Phi = \begin{bmatrix} \phi(\mathbf{x}_1)^\top \\ \vdots \\ \phi(\mathbf{x}_n)^\top \end{bmatrix} = \begin{bmatrix} \varphi_1(\mathbf{x}_1) & \dots & \varphi_M(\mathbf{x}_1) \\ \vdots & & \\ \varphi_1(\mathbf{x}_n) & \dots & \varphi_M(\mathbf{x}_n) \end{bmatrix}$ is called the **design matrix**.

MLE of the Weights

- Recall $\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}$, $\frac{\partial(\mathbf{x}^\top \mathbf{A} \mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}$.

- Taking gradient w.r.t. $\mathbf{w}$ and setting to zero,

$$\Phi^\top \Phi \mathbf{w}_{\text{ML}} - \Phi^\top \mathbf{y} = 0$$

$$\mathbf{w}_{\text{ML}} = (\Phi^\top \Phi)^{-1} \Phi^\top \mathbf{y}$$

if $\Phi \in \mathbb{R}^{n \times M}$ has **linearly independent** columns (need $n \geq M$, i.e., **more data samples than dimension of parametric space**). Also known as *least squares solution*.

Example

- $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.
- $\phi(\mathbf{x}) = [1, x_1, x_2, x_1^2, x_2^2]$, $y = w_1 + w_2x_1 + w_3x_2 + w_4x_1^2 + w_5x_2^2 + \epsilon$.

$$\varphi_1(\mathbf{x}) = 1, \quad \varphi_2(\mathbf{x}) = x_1, \quad \varphi_3(\mathbf{x}) = x_2, \quad \varphi_4(\mathbf{x}) = x_1^2, \quad \varphi_5(\mathbf{x}) = x_2^2$$

- $\mathbf{x}_1 = \begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}$, $\dots$, $\mathbf{x}_n = \begin{bmatrix} x_{n1} \\ x_{n2} \end{bmatrix}$.

$$\Phi = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{11}^2 & x_{12}^2 \\ 1 & x_{21} & x_{22} & x_{21}^2 & x_{22}^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n1}^2 & x_{n2}^2 \end{bmatrix}$$

Goodness of Fit

- Residual sum of squares $RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$.
- Root mean squared error $RMSE = \sqrt{\frac{1}{n} RSS}$.
- Coefficient of determination

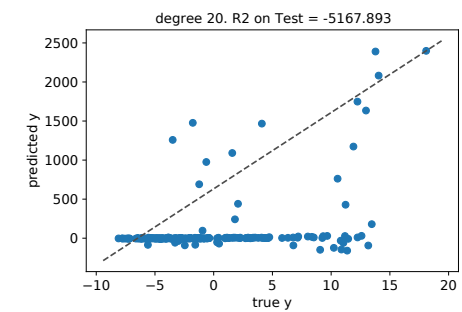
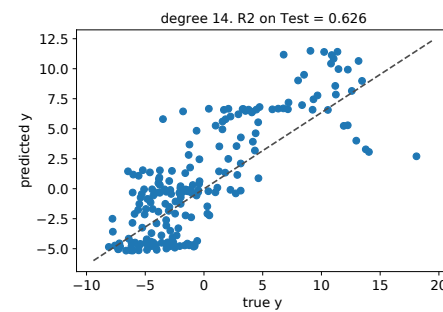
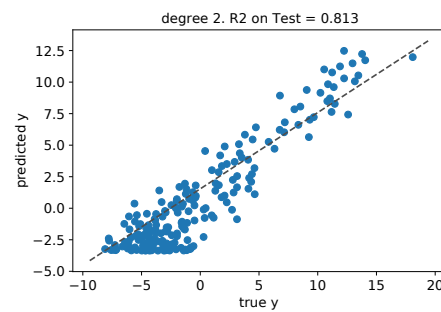
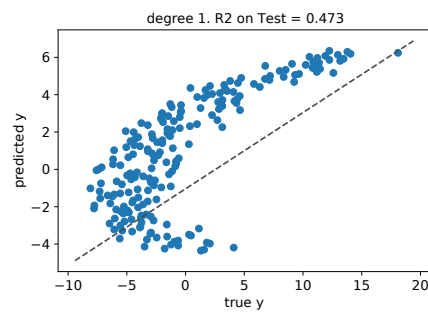
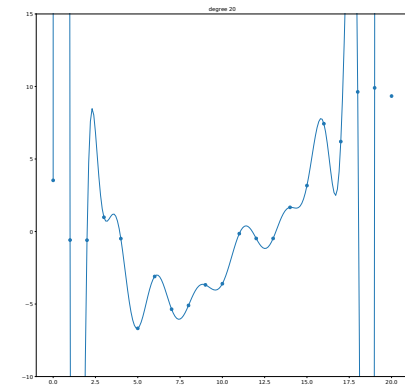
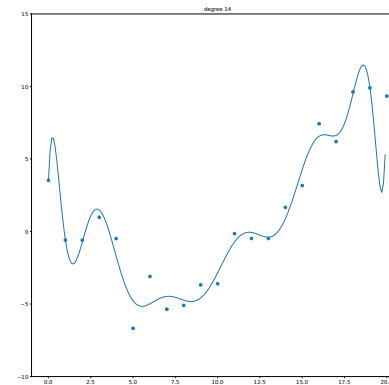
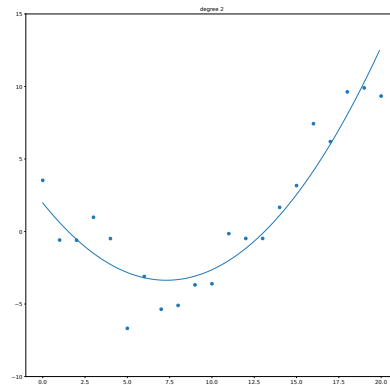
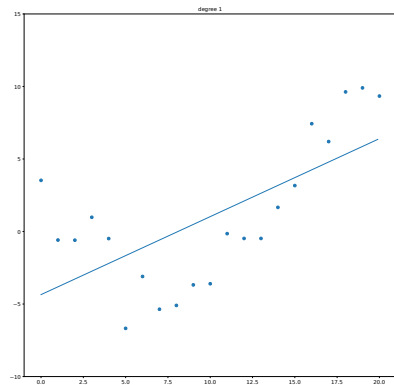
$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{RSS}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

where $\bar{y} = \frac{1}{n} \sum_i y_i$. TSS is the total sum of squares or the empirical variance of the data.

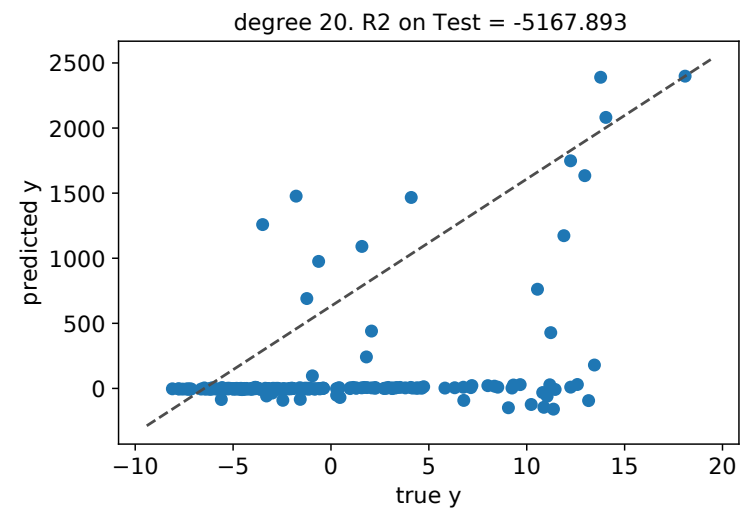
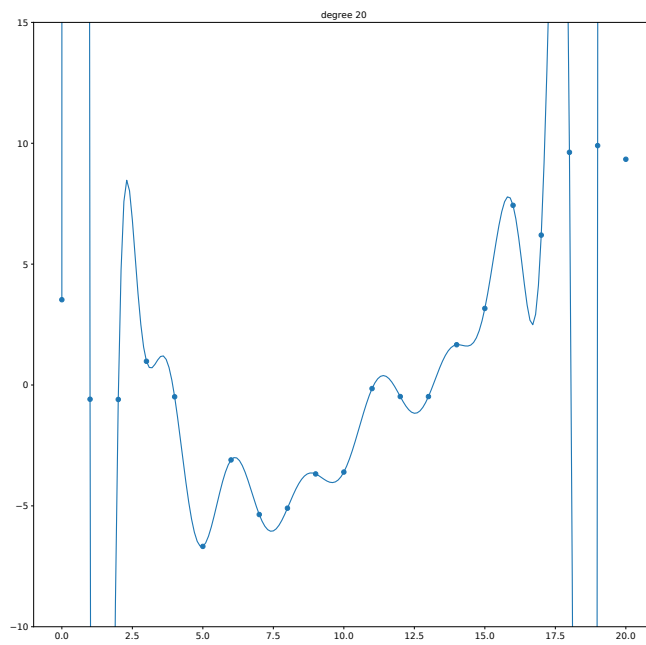
- ▶ R^2 measures the amount of variance in the data *explained* by the model with larger R^2 implying a better fit.
- ▶ E.g., if $\hat{y}_i = \bar{y}$ for all i (constant prediction using empirical mean), then $R^2 = 0$.
- ▶ If $R^2 < 0$, the model is worse than constant prediction!

Fitting Different Degrees

02_linreg_poly_vs_degree.ipynb



Fitting Different Degrees



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Maximum A Posteriori (MAP) Estimate

- In some applications, we may have **prior** knowledge of the parameter distribution.
- MAP estimate of a model's parameter θ :

$$\theta_{\text{MAP}} = \arg \max_{\theta} \underbrace{p(\theta \mid \mathcal{D})}_{\text{posterior}}$$

- Taking log:

$$\begin{aligned} p(\theta \mid \mathcal{D}) &\propto p(\mathcal{D} \mid \theta)p(\theta) \\ \log p(\theta \mid \mathcal{D}) &= \log p(\mathcal{D} \mid \theta) + \log p(\theta) + \text{const.} \end{aligned}$$

MAP Example

Suppose $p(\mathcal{D} \mid \theta) = \theta^{N_1} (1 - \theta)^{N_0}$, where $N_k = \sum_{i=1}^n \mathbf{1}\{x_i = k\}$. The prior of θ is $p(\theta) = \text{Beta}(\theta \mid a, b)$. Find the MAP estimate of θ .

$$\log(p(\mathcal{D} \mid \theta)p(\theta)) = (N_1 + a - 1) \log \theta + (N_0 + b - 1) \log(1 - \theta).$$

Taking derivatives w.r.t. θ and setting to zero:

$$\begin{aligned} \frac{N_1 + a - 1}{\theta_{\text{MAP}}} &= \frac{N_0 + b - 1}{1 - \theta_{\text{MAP}}} \\ \theta_{\text{MAP}} &= \frac{N_1 + a - 1}{n + a + b - 2}. \end{aligned}$$

Example: Classification

- Suppose each feature vector $\mathbf{x} \in \mathcal{X}$ has a class label $y \in \{1, \dots, K\}$.
- Decision rule $\delta(\mathbf{x}) = k$ if $\mathbf{x} \in \mathcal{R}_k$, $k = 1, \dots, K$. We want to determine the optimal decision regions $\mathcal{R}_1, \dots, \mathcal{R}_K$.
- For a given $\mathbf{x}$, its classification error is given by

$$\sum_{y \neq \delta(\mathbf{x})} \mathbb{P}(Y = y \mid \mathbf{x}) = 1 - \mathbb{P}(Y = \delta(\mathbf{x}) \mid \mathbf{x}).$$

- Therefore, to minimize the classification error, we let $\delta(\mathbf{x}) = k$ where $p(y = k \mid \mathbf{x})$ is the largest, i.e., we take the MAP estimate of y given x .

Example: Naive Bayes

- Given the image $\mathbf{x}$, what digit is it?



- Assume that the features in $\mathbf{x} = (x_1, \dots, x_D)$ are conditionally independent given the class label y .

$$p(\mathbf{x} \mid y = c, \boldsymbol{\theta}) = \prod_{d=1}^D p(x_d \mid \boldsymbol{\theta}_{dc}),$$

where $\boldsymbol{\theta}_{dc}$ are the parameters for the class c and feature d .

- Let π be the prior probability distribution of y . Posterior distribution of y :

$$p(y = c \mid \mathbf{x}, \boldsymbol{\theta}) \propto \pi(c) \prod_{d=1}^D p(x_d \mid \boldsymbol{\theta}_{dc})$$

Example: Naive Bayes

- Binarized version of MNIST (<http://yann.lecun.com/exdb/mnist/>) serves as the features $\mathbf{x}$. The class label y is the digit the image belongs to.
- Binary features, $x_d \in \{0, 1\}$: $x_d \mid y = c \sim \text{Bern}(\theta_{dc})$.
- Simple model but test accuracy 84.3%!

https://colab.research.google.com/github/probml/probml-notebooks/blob/master/notebooks-d21/naive_bayes_mnist.ipynb

Summary

- Principles of Bayesian inference.
 - ▶ MLE
 - ▶ MAP

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Some Practice Problems

1. Suppose $x_1, \dots, x_n \sim \mathcal{N}(\cdot \mid \mu, \sigma^2)$ i.i.d. with known variance σ^2 . Find the MLE of μ .

Solution: The log likelihood is

$$\log p(\mathcal{D} \mid \mu) = \log \prod_{i=1}^n \mathcal{N}(x_i \mid \mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 + \text{const.}$$

The first order derivative w.r.t. μ is

$$\frac{d}{d\mu} \log p(\mathcal{D} \mid \mu) = -\frac{1}{2\sigma^2} \sum_{i=1}^n 2(x_i - \mu)(-1) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu).$$

Some Practice Problems (cont.)

Finding the root of the first order derivative, we have

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu_{\text{ML}}) = 0$$

$$n\mu_{\text{ML}} = \sum_{i=1}^n x_i$$

$$\mu_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Some Practice Problems (cont.)

2. Continuing from the previous question, suppose that the prior distribution of μ is $\mathcal{N}(\cdot | \mu_0, 1)$. Find the MAP estimate of μ .

Solution:

$$\begin{aligned}\log p(\mu | \mathcal{D}) &= \log p(\mathcal{D} | \mu) + \log p(\mu) + \text{const} \\ &= -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 - \frac{1}{2}(\mu - \mu_0)^2 + \text{const}.\end{aligned}$$

The first order derivative w.r.t. μ is

$$\begin{aligned}\frac{d}{d\mu} \log p(\mu | \mathcal{D}) &= -\frac{1}{2\sigma^2} \sum_{i=1}^n 2(x_i - \mu)(-1) - (\mu - \mu_0) \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) - \mu + \mu_0.\end{aligned}$$

Some Practice Problems (cont.)

Finding the root of the first order derivative, we have

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu_{\text{MAP}}) - \mu_{\text{MAP}} + \mu_0 = 0$$

$$\mu_{\text{MAP}} = \frac{\frac{1}{\sigma^2} \sum_{i=1}^n x_i + \mu_0}{\frac{n}{\sigma^2} + 1}.$$

3. Explain each line of the following code snippet.

```
1 model = LinearRegression()  
2 poly_features = PolynomialFeatures(degree=d, include_bias=True)  
3 Xtrain_poly = poly_features.fit_transform(Xtrain)  
4 model.fit(Xtrain_poly, ytrain)
```

Solution:

Line 1: Creates a Linear Regression object.

Some Practice Problems (cont.)

- Line 2: Creates a polynomial features class of degree d with bias term.
- Line 3: Generates the training feature matrix.
- Line 4: Fits the linear regression model.